

Roger Cohen

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SENIOR DATA ANALYST SUMMARY

- 10+ years of experience turning complex datasets into actionable insights through quantitative research, Python-based analytics and data visualization in agile, cross-functional environments.
- Proven leader of 15+ data projects, securing and managing \$650K+ in competitive federal funding.
- Skilled communicator with a record of translating technical findings for diverse audiences: author of 100+ peer-reviewed journal articles (h-index=37) and presenter at 30+ international conferences.

KEY SKILLS

- **Programming & Data Analysis:** Python (NumPy, SciPy, matplotlib, pandas, scikit-learn, PyTorch, sbi, emcee), SQL, Excel
- **Statistical & Machine Learning Methods:** Likelihood-based inference, regression modeling, experimental design (hypothesis, A/B testing), dimensionality reduction (PCA), forward modeling (MCMC, monte carlo)
- **Tools & Environments:** Git, Google Cloud, AWS, Slurm (HPC queue scheduler), Jupyter, Linux/Unix, Windows 365, MacOS

EXPERIENCE

Senior Data Analyst | Rutgers University | Piscataway, NJ (hybrid/remote) | Sep 2021 – Present

- Developed and deployed automated Python pipelines leveraging Slurm-based HPC workflows to process 500+ observations of 111 targets, reducing processing time by 80%.
- Implemented Bayesian maximum-likelihood models to fit noisy, asymmetric data across 40 datasets, improving parameter precision and reliability.
- Managed >1 TB of structured data across 250+ datasets, optimizing accessibility and analysis efficiency.
- Mentored and managed four junior data analysts in Python-based data wrangling, statistical modeling and visualization, scoping and delivering end-to-end analytical projects.

Subject Matter Expert: Data Science | SweetRush | San Francisco, CA (remote) | Sep 2025

- Beta tested and evaluated four advanced data science/data engineering courses for a \$1.6B corporate client
- Collaborated with instructional designers to enhance technical accuracy, clarity and real-world applicability.

Senior Data Analyst | Space Telescope Science Institute | Baltimore, MD | Nov 2016 – May 2021

- Built and led an international team of 6 data scientists, delivering a \$200K grant-funded project ahead of schedule and under budget.
- Led five end-to-end data projects in parallel, driving publication in top-tier journals and generating 100+ citations and earning regular media coverage.
- Designed Python-based data visualizations, communicating project progress and outcomes in 20+ presentations across 3 continents.
- Constructed a pipeline for 2D data smoothing and interpolation across 16 large datasets (>10K samples each), achieving a >20% reduction in processing time.

Data Analyst | Universidad de Concepción | Concepción, Chile | Aug 2012 – Oct 2016

- Automated SQL-Python data pipelines to query, extract and cross-match 10M+ records, cutting manual processing time by >50% and reducing human error.
- Implemented weighted least-squares fitting with sigma clipping through a custom analysis workflow, improving runtime performance by >15%
- Held key leadership role in 100+ member international data consortium, tracking performance metrics and forecasting project efficiency, leading to >10% improvement in throughput.
- Managed cross-functional communication among 15+ global collaborators, leading weekly meetings for three years and ensuring alignment across distributed teams.

EDUCATION

Ph.D., M. Sc., Astronomy

B.A., Physics and Astronomy (honors)

University of Florida, Gainesville, FL

Wesleyan University, Middletown, CT